

Table 4: Effect of LLM exposure on occupation-level job postings

|                             | Job postings (log)  | Salary                   | Salary (log)         | Salary (SD)          |
|-----------------------------|---------------------|--------------------------|----------------------|----------------------|
| LLM exposure x Post         | -0.039**<br>(0.018) | -495.483***<br>(135.270) | -0.011***<br>(0.003) | -56.802<br>(108.358) |
| Occupation fixed-effects    | Yes                 | Yes                      | Yes                  | Yes                  |
| Year-by-month fixed effects | Yes                 | Yes                      | Yes                  | Yes                  |
| Observations                | 16,815              | 16,815                   | 16,815               | 16,804               |
| Adjusted R <sup>2</sup>     | 0.962               | 0.948                    | 0.953                | 0.729                |

*Notes:* This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level outcomes. *Job postings (log)* is the log number of new vacancies in occupation  $o$  in month  $t$ . *Salary* is the mean advertised annual compensation, *Salary (log)* is its logarithm, and *Salary (SD)* is the within-occupation standard deviation of posted salaries. *Exposure* is the occupation-level LLM exposure score from Eloundou et al. (2024). *Post* equals one for November 2022 onward. Standard errors are clustered at the occupation level. Significance is denoted by \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

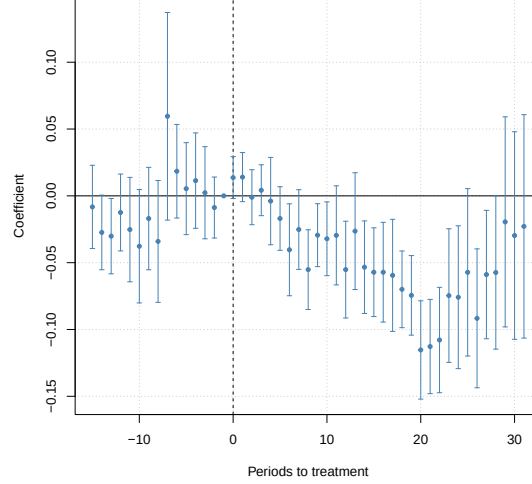
#### 4.2.2 Occupation-Level Postings

This analysis examines occupation-level job posting data to understand how aggregate labor demand evolves for specific roles. This analysis exploits variation in LLM exposure across 384 occupations, from minimally affected roles such as cooks (exposure score: 0) to highly exposed positions like proofreaders and copy markers (exposure score: 0.86). We examine three outcomes: posting volumes to measure aggregate demand, advertised salaries to assess compensation responses, and within-occupation salary dispersion to measure wage polarization.

Figure 6 presents event study estimates for job posting volumes. Prior to November 2022, high- and low-exposure occupations follow parallel trajectories, validating our identification strategy. Following ChatGPT’s release, however, exposed occupations experience a sustained reduction in hiring demand, with effects first becoming statistically significant after 7 months and intensifying thereafter. After 20 months, we see a slight resurgence, and job postings converge back to their original levels. When averaged across the post-treatment period (Table 4), a one standard deviation increase in LLM exposure is associated with a 3.9% reduction in posting volume ( $p = 0.029$ ). This is an economically meaningful change.

Turning to compensation, we observe immediate and persistent wage pressure for exposed occupations. Figure 7 indicates that posted salaries begin declining shortly after ChatGPT’s introduction. A one standard deviation increase in LLM exposure is associated with a £495 decline in posted salaries and a 1.1% reduction in log

Figure 6: Effect of LLM exposure on job-level hiring



*Notes:* This figure plots event study estimates of the effect of LLM exposure on occupation-level hiring (log vacancies per occupation-month). Coefficients are estimated from Equation (5) with occupation fixed effects and year-by-month fixed effects; standard errors are clustered at the occupation level. The dependent variable is the log of the number of new vacancies posted in occupation  $o$  in month  $t$ . Exposure is the occupation-level score from Eloundou et al. (2024). The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

salaries (both  $p < 0.001$ ). Panel C shows that within-occupation salary dispersion remains unchanged, suggesting uniform wage pressure across the distribution rather than polarization between high- and low-earners within occupations.

These occupation-level salary declines provide a sharp contrast with the slight increase in firm-level compensation documented earlier. At the firm level, eliminating junior positions mechanically raises average compensation by shifting workforce composition toward senior roles. At the occupation level, however, we observe market-wide wage depression for exposed roles as increased labor supply (from displaced workers) meets reduced demand. As such, our results highlight the importance of considering both within-firm and within-occupation changes.